Bounce-apart calculations

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use momentum and energy to check an elastic collision.

SUCCESS CRITERIA - I can...

1. Write a signed momentum balance.
2. Check kinetic energy separately.
3. Reject an elastic claim when kinetic energy changes.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “elastic” and “conserved” to explain your model or result. Sentence starters are optional.

elastic: a collision that conserves total kinetic energy

conserved: unchanged in total under the stated conditions

kinetic energy: energy associated with motion

01 Warm-up: bring back an earlier idea

Two 3 kg carts latch together after colliding. Is this perfectly inelastic?
State what remains conserved if external impulse is negligible.

Word help: inelastic = a collision that transforms some kinetic energy to other forms

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p_{total,\,i} = p_{total,\,f}$; $K_{total,\,i} = K_{total,\,f}$.
Isolated elastic collision; $K = \frac{1}{2}mv^2$. Equal masses exchange velocities in the ideal one-dimensional case.

Worked reasoning: A 1 kg cart moving at +3 m/s hits a stationary 2 kg cart. The first rebounds at -1 m/s. Total momentum was +3 kg·m/s, so the second must have momentum +4 kg·m/s and velocity +2 m/s.

Your turn: Draw momentum arrows before and after; verify the total.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Equal-mass carts collide head-on elastically. A starts right at 3 m/s; B rests.
Sketch the after state.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Before contact: add velocity arrows.



Marzano check - circle: 1 2 3

04 Apply the physics

A 1 kg cart approaches a bumper at 5 m/s right and rebounds at 1 m/s left. Take right as positive.
Label both velocities and calculate the cart's momentum change.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A pair's total momentum matches before and after, but kinetic energy falls. Is the collision elastic?
Explain 0.

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

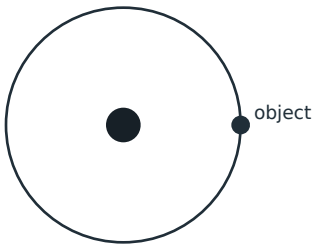
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A satellite follows a circular orbit at a constant 6 km/s.
Draw velocity and acceleration at the top and side of the orbit.
Explain the different directions.
Label velocity and inward acceleration on the circular model.

Word help: orbit = a path around another body under gravity

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Equal-mass carts have initial velocities +4 and -1 m/s. In an ideal 1 D elastic collision, what are final velocities? Organize the known quantities on the before-and-after model.

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice … . This supports … because … .

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 2 kg cart starts at 3 m/s and strikes a stationary 1 kg cart. If the heavy cart finishes at 1 m/s right, find the light cart's velocity. Organize the known quantities on the before-and-after model.

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3